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RESEARCH

Temporal trends and spatial variations of total mercury in sediments of the legacy-contaminated river Elbe (Germany)

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Abstract River systems often retain the imprint of historical mercury (Hg) contamination over long time scales since decreasing trends after reduced Hg inputs are impeded by legacy storage in sediments. Here, we report temporal and spatial variations of total mercury (THg) concentrations in sediments of the Elbe River, Germany. Temporal THg trends (1996–2019) were investigated at five long-term monitoring sites in harbors and groyne fields. In contrast to well-documented declines of THg in suspended particulate matter (SPM) in recent decades, significant trends of decreasing THg in sediments were only found for two harbors, while the other sites exhibited no decreasing trends. Spatial distribution patterns of THg in sediments (0.11–10.6 mg kg^{-1}) showed increases in areas of the Middle Elbe downstream of the confluences of the tributary rivers Mulde and Saale. Similar spatial patterns of increasing concentrations were observed in floodplain soils (0.04–4.20 mg kg^{-1}) and SPM (0.37–1.84 mg kg^{-1}). Regardless of the spatial origin or sample type (sediments, floodplain soils, SPM), the binding forms of Hg studied by sequential extraction showed a clear dominance of strongly bound Hg extractable with 12 M HNO_3. While this seems to indicate a low mobility of Hg, uptake into aquatic food webs may occur following methylation. Our results demonstrate that the reduction of Hg releases into the Elbe River basin has so far only been able to improve sediment quality in some parts. Deposits of Hg-containing sediments will therefore remain a future problem with implications for the continued exceedance of environmental quality standards and the management of waterways.

Keywords Hg · Elbe river · Suspended particulate matter · Floodplain soils · Trend analysis · Sequential extraction

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Introduction

As a threat to biodiversity and human health, the contamination of streams by mercury (Hg) is of global concern (Randall & Chattopadhyay, 2013; Peng et al., 2025). In streams and floodplains, Hg can be taken up directly by organisms or enter food webs after conversion to highly toxic and bioaccumulative

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methylated Hg species due to microbial methylation under anoxic conditions (Bravo & Cosio, 2020). Although the industrial use of Hg and the production of Hg-containing commercial products is declining since the late twentieth century and the emissions of Hg decreased in many regions over the past decades (Horowitz et al., 2014; Geyman et al., 2025; Qiu et al., 2025), river systems are often still impacted by Hg as a legacy of historic inputs (e.g., Dickson et al., 2019; Lazareva et al., 2019; Biester et al., 2000; Hošek et al., 2020; Rudd et al., 2021). In Germany, for example, as a long-term consequence of historical mercury inputs, the environmental quality standard for Hg in biota (20 µg/kg wet weight) is exceeded in almost all watercourses (BMUV/UBA, 2022). Hg-enriched sediments represent a potential threat for water quality and river ecosystems, since they can be remobilized during flood events or engineering works (Ciszewski & Grygar, 2016). Therefore, knowledge of the temporal and spatial variability of sediment contamination at the catchment level is of great importance in the context of water engineering or river restoration in metal-contaminated waterways such as the Elbe (Germany), one of the historically most metal-polluted rivers in Europe (Förstner et al., 2004). Particularly in areas of the non-tidal Elbe, Hg is comparatively the most problematic pollutant for achieving the river management objectives (Heise et al., 2008).

Several studies documented decreasing trends of Hg concentrations in suspended particulate matter (SPM) of the Elbe during the past 30 years (Schwartz et al., 2015; Wellmitz, 2015). Temporal trends of river-specific trace elements are usually recorded during official monitoring programs by analyzing samples from water (Romero et al., 2016) or SPM (Wang et al., 2019), while investigations on the temporal development of Hg in Elbe sediments were less frequently conducted. Prange et al. (2000) and Heininger et al. (2003) found decreasing concentrations of Hg in sediments between 1992–1998 and 1991–2001, respectively. In contrast, Heise (2015) showed that strongly elevated Hg concentrations are still prevailing in sediments of Elbe river backwaters. It therefore remains uncertain whether the recent decline in Hg releases into the river is reflected in long-term trends of sediment contamination. Based on this, we decided to test as our first research hypothesis that time series of Hg concentrations in fine sediments of harbors and groyne fields do not show decreasing trends in the recent past, in contrast to the documented decreasing trends in SPM (Fliedner et al., 2014; German Environment Agency, 2025).

Time series of Hg in sediments are often based on the analysis of sediment cores (Wentz et al., 2014). In waterways heavily influenced by hydraulic engineering such as the Elbe river, however, unaltered sediment archives are often difficult to find and verify and, thus, especially in the case of undisturbed fine sediment deposits, relatively rare. This is due to the fact that fine sediment sinks are mostly located in anthropogenic operating areas of the river system (i.e., harbor basins, groyne fields), where their layers are usually disturbed by shipping, dredging, or construction works. These circumstances often make it difficult to analyze the temporal development of Hg concentrations on the basis of sediment profiles. The long-term analysis of disturbed fine surface sediments sampled consecutively from low-flow areas at a constant geographical position can be a viable alternative. However, this approach requires regular sampling over many years, as disturbed samples may mix recent sediment deposits with older ones, so that changes in sediment quality over time can usually only be detected after many years of sampling. In addition, long-term analysis of disturbed sediment samples can be affected by small-scale variability in sediment composition, which may be erroneously interpreted as temporal changes of Hg concentrations since local heterogeneities can often not be sufficiently considered during sampling. The analysis of multiple sediment layers is not possible using disturbed grab samples from the sediment surface. However, this type of sampling is usually conducted for sediment assessment in the context of waterway maintenance, so that it provides very practice-oriented results. The short time required to take disturbed sediment samples enables large numbers of samples to be collected. This enables the investigation of accompanying questions, for example, regarding the spatial distribution of Hg within a study area, which can be used additionally to interpret time series data from multiple sampling campaigns in different years.

In particular, sediments from river sections with low flow velocity and floodplain soils (FS) can contain high proportions of legacy deposits (Singer et al., 2013; Lecce & Pavlowsky, 2014; Anatone et al., 2020; Heaven et al., 2000). SPM, rather, derives from erosion products of younger riverine material (Viers

et al., 2009; Zeng et al., 2022). It is more representative of the Hg amount which is currently transported in the river and may therefore be comparatively more reactive to temporal changes in Hg inputs to the river system than sediments. Although the spatial distribution of Hg is well documented individually in sediments (Wallschläger et al., 1998a; Prange et al., 2000; Heise et al., 2005), FS (Krüger & Gröngröft, 2003), and SPM along different sections of the non-tidal Elbe, there is little knowledge about the extent to which Hg contamination and mobility differ between potential aquatic (fine sediments) and (semi-)terrestrial (FS) pollutant sinks and the particulate load currently transported in the river, represented by SPM. The currently available data sets on THg in the Elbe do not allow us to draw any conclusions on its mobility and potential bioavailability (Issaro et al., 2009). Furthermore, the binding forms of Hg may vary depending on its origin (Fernández-Martínez et al., 2019), and the Elbe River has been exposed to the influence of various Hg sources in the past. This led to our second research hypothesis that sediments differ significantly in their binding behavior both along the length of the Elbe and in relation to FS and SPM. To address both hypotheses, the primary goals of this manuscript were to combine a study on long-term trends of Hg in sediments at representative locations with investigations of Hg concentrations and binding forms in sediments, FS, and SPM along the non-tidal German part of the river Elbe.

Material and methods

Study region

On German territory, the Elbe River is divided into a tide-affected (downriver km 586) and a non-tidal influenced (km 0–586) section (Fig. 1). For this study, investigations were carried out in the non-tidal section between river-km 13 (site "Prossen", Saxony) and river-km 585 (site "Geesthacht," Schleswig–Holstein). Within this study region, the morphology of the Elbe basin changes significantly due to the transition of a low mountain landscape characterized by Palaeozoic (granite, granodiorite), Mesozoic (sand-, silt-, claystone), and Tertiary rocks into lowlands dominated by unconsolidated Quaternary sediments downriver of the city of Meißen/Saxony (Walter, 2007). The hydromorphology of the Elbe was anthropogenically modified by the removal of islands, the straightening of meanders, the development of dykes, and the construction of extensive groyne fields, while the active floodplains are mostly used as meadows or pastures (Rommel et al., 2012; Schulz-Zunkel, 2014). The catchment area of the Elbe River was subject to large Hg inputs during the twentieth century, whereby the releases from industrial point sources strongly decreased since the beginning of the 1990s. On German territory, the largest historical emitters were chlor-alkali and acetaldehyde plants at Buna and Bitterfeld, whose releases mostly reached the Elbe River via the tributaries Saale and Mulde. Even today, the largest inputs of Hg into the Elbe occur via these two tributaries, albeit to a much lesser extent than in the past. This is mostly due to diffuse inputs from contaminated sites in the tributary catchments and the periodic remobilization of contaminated old sediments (Schwartz et al., 2015).

Fieldwork

The field work included the regular sampling (1996–2019) of five long-term monitoring sites, which were used to analyze temporal variations of THg concentrations in sediments. To investigate spatial differences in THg and Hg binding, the sediment sampling campaigns were extended between 2017 and 2019 (Fig. 1). This dataset was additionally used to compare the enrichment and binding forms of Hg in sediments (SD) with floodplain soils (FS) and suspended particulate matter (SPM) of the Elbe river.

Investigations on temporal variations of THg

For the implementation of trend analyses for THg, sediments of groyne fields were regularly sampled at the sites "Elster" ($n = 17$) and "Hitzacker" ($n = 21$). Samples of the sites "Prossen" ($n = 16$), "Dessau" ($n = 22$), and "Geesthacht" ($n = 19$) were taken from the entrances of inland ports. Annual sampling campaigns during summer months were carried out between 1996 and 2019, although the sampling periods were somewhat variable between the different sampling sites (Table S1). At each sampling location, consistent coordinates were used as the target point. Sediments were sampled by means of a Van Veen grab sampler, which excavated sediments of ~ 0–15

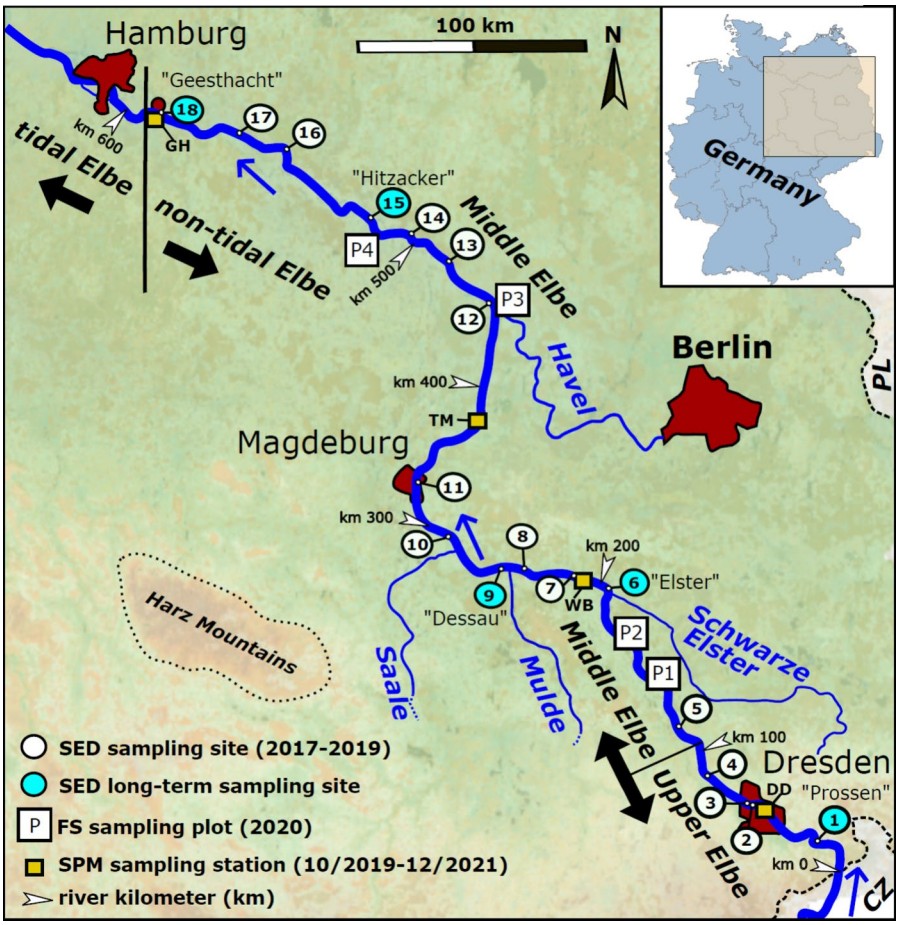

Fig. 1 The sampling sites of sediments (SED), FS, and SPM within the study area (Map based on BGR, 2023). The numbers of long-term sediment sampling sites correspond to the following sites and sampling periods: 1, Prossen (2002–2019); 6, Elster (2000–2018); 9, Dessau (1996–2019); 15, Hitzacker (1996–2018); 18, Geesthacht (1996–2018). Numbers 1–18 mark the sites of the sampling campaigns between 2017 and 2019. The abbreviated labels of the SPM sampling stations refer to Dresden (DD), Wittenberg (WB), Tangermünde (TM), and Geesthacht (GH)

cm depth. All samples were manually homogenized, stored in a refrigerator at 4 °C, and promptly freeze-dried after a sampling campaign was finished. The dried samples were homogenized again, sieved (< 2 mm), and milled by a planetary ball mill. The concentration of total nitrogen (N_t) was analyzed from three replicates of a respective sample by a CNS analyzer (vario Macro cube, Elementar) after combustion at 950 °C according to DIN (2012). The analysis of organic carbon (TOC) was conducted by a similar procedure by means of a TOC analyzer (Rapid CS cube, Elementar), after inorganic carbon was removed by the addition of HCl (10%) (DIN, 2022).

Total Hg (THg) was analyzed using a direct thermal decomposition Hg-analyzer (MA-3000, Nippon Instruments Corporation), in which the samples were heated to 600 °C and THg was determined by atomic absorption spectrometry after amalgamation. A recovery of 100 ± 10% was regularly controlled by the analysis of a certified reference material (NIST 2709). The total concentrations of aluminum (Al), silicon (Si), iron (Fe), manganese (Mn), and sulfur (S_t) of the milled samples were analyzed as loose powders by energy dispersive X-ray fluorescence spectrometry (XRF) (Xepos He, Spectro Analytical Instruments), accompanied with regular analyses of reference materials (NIST 2711; NCS DC 73014).

Investigations on spatial variations of THg and Hg binding forms

To study the spatial variation of THg and Hg binding fractions, sampling was extended between the years 2017 and 2019 to include a higher number of sampling locations (Fig. 1, sampling locations 1–18, Table S2), which all comprised harbor entrances and groyne fields, including the long-term sampling sites. Sampling was conducted as described above.

To compare Hg concentrations and binding forms in sediments with those in FS and SPM, the sample set was supplemented by samples from FS, which were collected in 2020 during the "El-Biota project," managed by the Helmholtz Centre for Environmental Research, in areas of 4 soil plots, situated at Elbe km 167.5–168.5 (site 1), km 191.0–191.5 (site 2), km 437.0–438.0 (site 3), and km 512.5–513.0 (site 4). Within each soil plot, a minimum of three soil profiles situated in floodplains at a distance of 1–15 m from the river channel was sampled, totaling 13 soil profiles. Soil samples were taken by means of an Edelmann auger. In order to record younger, near-surface and older, deeper deposits while maintaining the feasibility of the investigations, sampling was limited to increments of 0–10 cm and 50–60 cm depth at each profile. The aim of this approach was to obtain a heterogeneous set of samples from comparable locations in the floodplain that reflects the variability of soil contamination as realistically as possible, even with a limited number of samples.

In order to compare the concentrations of "classical" Hg sinks (sediments, FS) with the current contamination status of the Elbe, SPM samples were additionally collected. Sampling of SPM was conducted at the monitoring stations DD (Dresden, km 57), WB (Wittenberg, km 216), TM (Tangermünde, km 388), and GH (Geesthacht, km 583), which are part of the monitoring network of the Federal Institute of Hydrology. Hg concentrations in SPM can vary over a year (Saniewska et al., 2014). Therefore, a comparatively high number of samples were collected to minimize the influence of seasonal effects. Sampling was conducted monthly by means of continuously operated cylindric sedimentation tanks (see Harhash et al., 2023 for details) between Oct. 2019 and Dec. 2021. An overview about the different sample types is provided in Table 1.

For laboratory analyses, all samples were prepared as described above. However, samples were aliquoted before they were milled, so that non-milled samples could be analyzed for grain size distribution. For this purpose, 0.5–1 g were mixed with 75–150 ml deionized water, and 1 ml of $Na_4P_2O_7$ (4%) was added. After overnight exposure, the suspensions were sonicated for 10 min. Grain size distributions were subsequently determined by laser diffractometry (Partica LA-960V2, Horiba Scientific).

Table 1 Overview about sample types, investigation periods, and sampling methods

Aim: Study on long-term trends of Hg in sediments at representative locations			
Sediment			**Analyses**
1996–2019: Repeated sediment survey at long-term monitoring stations of the Elbe; Van-Veen-grab (5 locations, 95 samples).			THg, TOC, N_t, S_t, Al, Si, Fe, Mn.
Aim: Investigation of the spatial variability of THg and Hg mobility			
Floodplain soil	**Sediment**	**SPM**	**Analyses**
June 2020; Soil investigation in areas of 4 study sites; Edelmann auger (25 soil samples).	2017–2019; Repeated sediment survey in the longitudinal course of the Elbe; Van-Veen-grab (18 locations, 38 samples).	Nov. 2019 – Dec. 2021; Repeated collection of SPM samples; sedimentation tanks (4 locations, 61 samples).	THg, Hg binding forms (sequential extraction), TOC, N_t, S_t, Al, Si, Fe, Mn, grain size distribution.

Since small-scale variations in sediment properties can significantly influence the concentration of a sediment contaminant, studies on the spatial distribution of contaminants are frequently carried out on the sediment fine grain fractions < 63 µm or < 20 µm (Dendievel et al., 2022). In Germany, however, numerous regulations relevant to sediment management refer to the < 2 mm fraction in sediments or FS. To address scientific questions and to collect data simultaneously relevant to practical applications, we decided to analyze the < 2 mm fraction and, to enable a trend analysis, to normalize THg concentrations by calculating the THg/TOC ratio. The selection criteria for this method were, on the one hand, the fact that THg and TOC showed higher correlation coefficients at all long-term measurement sites than the elements Al and Fe (Table S3), which might also be used for normalization. Furthermore, large proportions of THg in aquatic ecosystems are associated with organic material; thus, the THg/TOC ratio is a particularly suitable normalization tracer for anthropogenic Hg contamination of top soils and sediments (Xue et al., 2019).

The concentrations of THg, aluminum (Al), silicon (Si), iron (Fe), manganese (Mn), sulfur (S_t), TOC, and total nitrogen (N_t) in milled samples were quantified as described above. Hg binding forms were analyzed using the sequential extraction method according to Bloom et al. (2003), which divides the Hg contents into five operationally defined fractions (F1 – F5). The extraction agents were ultrapure water (F1), a "stomach-acid" mixture of 0.1 M acidic acid and 0.01 M HCl (pH 2.0) (F2), 1.0 M KOH (F3), 12 M HNO_3 (F4), and aqua regia (F5). F1 to F4 were extracted by shaking the samples for 18 ± 4 h in the respective extracting agents. F5 was extracted after an equally long reaction in loosely capped glass tubes. After each extraction step, the extracts were separated from the solids by centrifugation (3000 g, 20 min) and decantation and transferred to a volumetric flask containing 1 vol-% bromine monochloride solution (prepared after Bloom et al., 2003). Each sample was then washed by shaking for 15 min with the extraction agent of the respective extraction step, which was added to the respective sample after repeated centrifugation. After online reduction by $SnC1_2$ and amalgamation enrichment by a gold trap, the Hg concentrations in solution were determined by cold vapor atomic fluorescence spectrometry (Mercur, Analytik Jena). A matrix-matched calibration was conducted for every extraction step. Based on a sample mass of 500 mg, the limit of quantification was 2 µg kg^{-1}. Each run of analyses was controlled by the standard reference material NIST 1641e, accepting a recovery of 100 ± 10%. The analytical performance of the whole sequential extraction was evaluated by replicate extractions ($n = 7$) of the standard reference material NIST-2709, for which the sum of F1 – F5 achieved a recovery of 90.5–118.4%. The recovery of a respective sample was additionally controlled by calculating the percentage of the sum of all fractions (F1 – F5) on the sample-specific THg concentration. The recovery (sumF1 – F5/THg × 100) for samples varied within the whole dataset between 71.3 and 119.2%.

Statistical analyses

All data were tested for a normal distribution by the Shapiro–Wilk test. Since several parameters, including THg concentrations, showed a non-normal distribution, trend analyses and significance tests were carried out using non-parametric methods. The results were interpreted as significant when reaching a p value < 0.05.

Trend analyses for THg concentrations of the long-term sampling sites (Prossen, Elster, Dessau, Hitzacker, and Geesthacht) were performed by R Studio, using the Mann–Kendall trend test.

The relationship between the sampling time (year) and the THg/TOC ratio was evaluated by the non-parametric Kendall's tau correlation coefficient. Trends were visualized as smoothed trend lines using the R-package "ggplot2."

Significance tests and correlation analysis were performed using IBM SPSS Statistics 26. The presence of significant differences between different sample groups was tested using the Kruskal–Wallis H test. Significantly different datasets were subsequently analyzed by a Dunn–Bonferroni post hoc test. To examine relationships between the different datasets, correlation analyses were conducted using the non-parametric Spearman rank correlation coefficient (r_s).

To evaluate the intensity of THg accumulation, the index of geo-accumulation (Mueller, 1986) was calculated, following the equation:

$$I_{geo} = log_2(\frac{M_s}{M_r * 1.5})$$

where M_s is the THg concentration of a sample and M_r is the element-specific background of Hg. To enable comparability of Hg enrichment within the Elbe river basin, the Elbe specific Hg background concentration (0.2 mg kg^{-1}) according to Wilken and Wallschläger (1996) was used as reference value for the calculation of the I_{geo}. To evaluate the accumulation state, the I_{geo} of each element and sample group was evaluated according to the following classification: $I_{geo} < 0$, uncontaminated (class 0); I_{geo} 0–1, uncontaminated to moderately contaminated (class 1); I_{geo} 1–2, moderately contaminated (class 2); I_{geo} 2–3, moderately to strongly contaminated (class 3); I_{geo} 3–4, heavily contaminated (class 4); I_{geo} 4–5, heavily to extremely contaminated (class 5); $I_{geo} > 5$, extremely contaminated (class 6).

Results and discussion

Temporal variations of long-term monitoring sites (1996 – 2019)

All samples generally consisted of silty sediments with slightly variable proportions of sand and clay. The sampling sites Elster, Hitzacker (groyne fields), and Geesthacht (harbor) were more variable in concentrations of Al (Fig. S1). Due to the suitability of Al as a proxy for grain size distribution (Dung et al., 2013), this indicates a higher variability of grain sizes within these sample series in comparison to Prossen and Dessau. TOC and N_t were on average highest at Dessau and Prossen and lowest at Elster (Table S4). The concentrations of S_t were significantly lower at the Prossen and Elster sites than at Dessau, Hitzacker, and Geesthacht. This phenomenon can be explained by an increased geogenic sulfate background in glacial sediments of the Middle Elbe catchment (Walter, 2007) and inputs of mining-related increased sulfate concentrations by the tributaries Mulde and Saale (Strauch et al., 2001; Reumann et al., 2022).

The concentrations of THg at sites located on the Upper Elbe (Prossen: 0.46–2.05 mg kg^{-1}) and the Upper Middle Elbe (Elster: 0.39–4.45 mg kg^{-1}) were comparably lower and less variable than at sites of the Middle (Dessau: 1.31 – 12.2 mg kg^{-1}; Hitzacker: 1.23–17.4 mg kg^{-1}) and the Lower Middle Elbe (Geesthacht: 1.51–4.85 mg kg^{-1}) (Fig. 2). This increase in THg concentrations in the longitudinal course of the Elbe is mainly determined by Hg inputs from the tributaries Mulde and Saale at river km 260 and km 291, respectively, which act as important Hg sources for the Elbe river (Heininger et al., 2003; Barborowski & Einax, 2016). However, it should be noted that the THg concentrations found are highly dependent on the prevailing sedimentation conditions. The representativity of the presented long-term monitoring sites is therefore limited to the respective harbor basin or groyne field and cannot be transferred to larger areas of the Elbe's longitudinal course.

The concentrations of TOC were less variable at the harbors of Prossen (4.98–6.47%) and Dessau (5.14–7.28%) than at the groyne field sites Elster (1.12–5.22%) and Hitzacker (1.26–7.84%) or at the harbor of Geesthacht (2.52–5.61%). Significant changes in TOC concentrations during the study period that could influence the THg normalization were only observed to a very limited extent at Prossen (Fig. S2). However, due to the close connection between THg and TOC (Wallschläger et al., 1998b; Yang et al., 2007), the normalization of THg concentrations by the TOC content was considered the most appropriate approach for conducting trend analyses. During the interpretation of the trend analyses, it should be considered that the comparability of the results might be slightly reduced by differences in the investigation periods, which started already in the year 1996 at the sites Dessau, Hitzacker, and Geesthacht, while sampling in Prossen and Elster began in 2002 and 2000, respectively. Temporal trends of decreasing THg/TOC ratios were found to be significant at the sites 1 Prossen (tau: −0.48, $p \leq 0.01$) and 18 Geesthacht (tau: −0.38, $p \leq 0.03$) (Fig. 2), while for the other long-term sampling sites, no significant trends were observed. The robustness of the results was further corroborated by the fact that trend analyses of THg concentrations without normalization by TOC content show similar patterns with an only very slightly reduced significance at Prossen (Table S5, Fig. S3).

The presence of decreasing Hg levels at Prossen and Geesthacht, as illustrated in Fig. 2, is consistent with previous studies on THg in sediments (Prange et al., 2000; Netzband et al., 2002; Stammerjohann, 2004), SPM (Schwartz et al., 2015; Wellmitz, 2015;

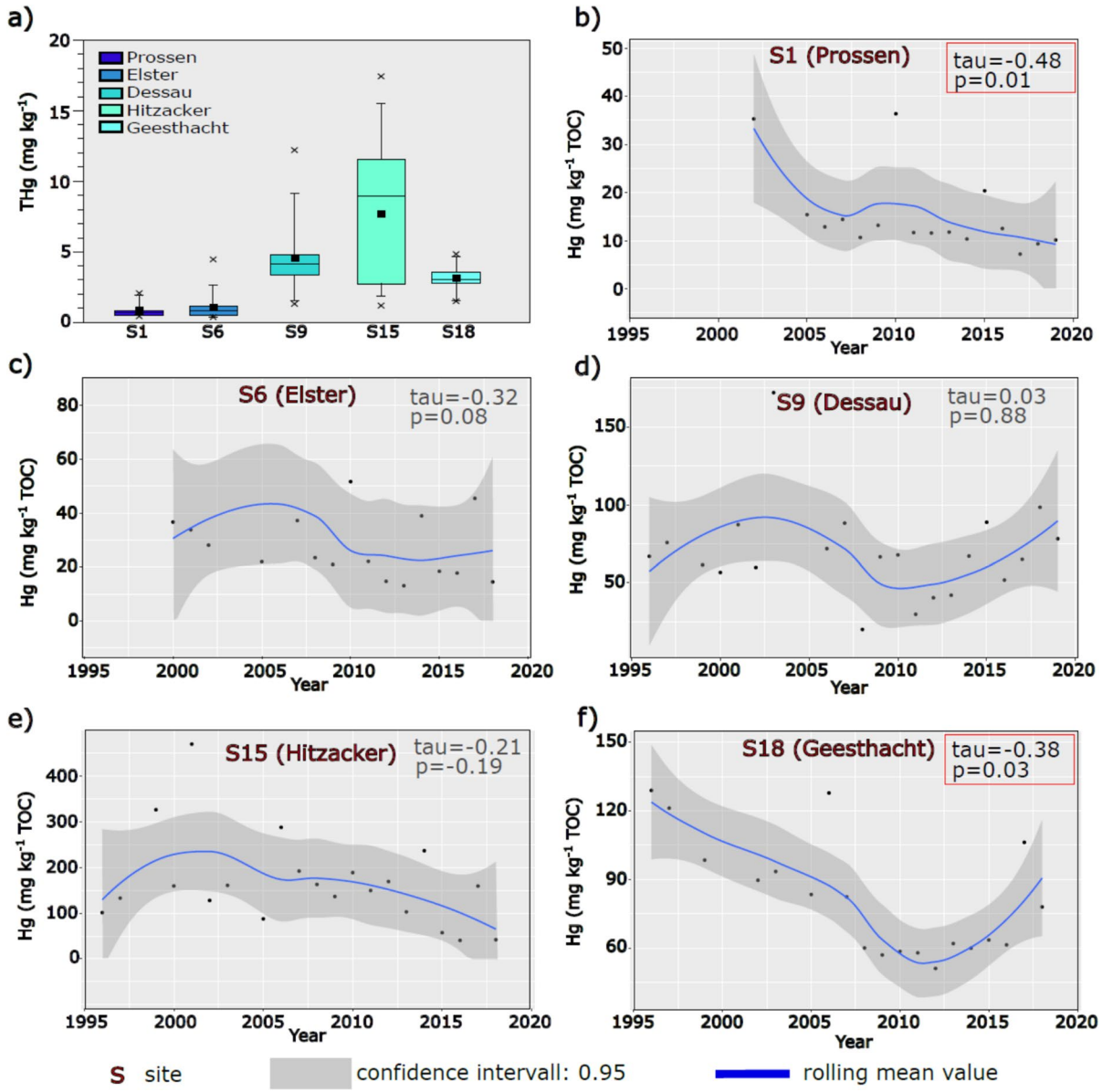

Fig. 2 Boxplots of THg concentrations (**a**) and trend diagrams (Mann–Kendall test) of THg/TOC ratio time series at the sites 1 (Prossen) (**b**), 6 (Elster) (**c**), 9 (Dessau) (**d**), 15 (Hitzacker) (**e**), and 18 (Geesthacht) (**f**). Significant trends ($p < 0.05$) are marked red

Krüger & Gröngröft, 2003), or biota (Kolaříková et al., 2012; Fliedner et al., 2014), which document a significant decline of metal inputs from point sources to the Elbe starting at the end of the 1980s. Schmidt et al. (2017) found decreases of 48–52% in the THg content in SPM from the Elbe at the locations Schmilka and Schnackenburg in a comparison of the periods 1990–2003 and 2004–2014. According to Wellmitz (2015), THg in SPM at Schmilka and Schnackenburg even decreased around 88% and 81%, respectively, between 1994 and 2013. The trends of decreasing THg concentrations in SPM found by both authors exceed the significant decreases of the THg/TOC ratio in sediments found in this study. For some harbors on the Elbe, Heininger et al. (2003) found strongly decreasing THg in sediments between 1991 and 2001, with concentration decreases occurring particularly during the early

1990s. We assume that this trend is not consistently present in the results of this study as the decline in concentrations was already significantly weakened by the start of the investigations in 1996, due to the earlier closure of major emitters. A similar legacy effect is described by Eklöf et al. (2024), who found a weakened decline of Hg in water of northern European rivers despite reduced Hg emissions.

In contrast to the studies cited above, it is remarkable that no significant decreases in the THg/TOC ratio of the long-term monitoring sites Elster, Dessau, and Hitzacker were found. Therefore, our first research hypothesis, that THg time series in sediments do not show decreasing trends, in contrast to SPM time series, could be confirmed for several locations. It can be assumed that sediments as potential contaminant sinks in harbors and groyne fields may need a longer time than SPM to be significantly affected by the declining Hg inputs into the Elbe. Furthermore, it is likely that sediments of the Middle Elbe are stronger subjects to the tributaries Saale and Mulde, which still contribute increased THg concentrations to the Elbe (Fliedner et al., 2014), especially during flood events (Baborowski & Einax, 2016). The tendency of a rather constant THg/TOC ratio between 1996 and 2019 at site Dessau likely reflects the influence of the Mulde River, whose THg concentrations in SPM even increased between 2000 and 2004 (Heise et al., 2005). In addition, Hg-enriched SPM from the Mulde could have been deposited in the harbor of Dessau during a large flood event in 2013, which would explain the increasing THg/TOC ratio there between 2014 and 2019. We assume that this influence on THg concentrations in sediments of the Elbe decreases with increasing distance from the confluences of the Mulde and especially of the Saale, from which the largest Hg load is discharged into the Elbe (Wilken & Hintelmann, 1991). However, a detailed quantification of the contributions of different historical emitters to Hg contamination of the Elbe and its tributaries did not lie in the scope of this study. Source tracing approaches using Hg stable isotope signatures (Wiederhold et al., 2015; Schwab et al., 2022) may potentially provide novel insights in future studies. At Hitzacker, where the widest THg/TOC ratio was found, no significant temporal trend could be recognized. Since a significant decreasing trend in the THg/TOC ratio was again observed for the entire investigation period at the downriver located site Geesthacht, it can be assumed that larger proportions of more heavily contaminated sediments were deposited in areas of the Middle Elbe (i.e., Dessau, Hitzacker) and thus did not yet reach this sampling site in higher amounts. This is consistent with the Hg concentrations in FS, which are highest in areas of the Middle Elbe between river kilometers 300 and 500 (Krüger & Gröngröft, 2003). However, although a significant decreasing trend was found for the site Geesthacht over the whole investigated period (1996–2018), THg rather increased during the last years (2012–2018) (Fig. 2b). Since a partially similar tendency was found in sediments of the German Bight (Kammann et al., 2023), the Hg concentrations in sediments at Geesthacht should be monitored in future studies. Even though the mentioned large flood event of 2013 may have affected the concentration increase, the reasons for increasing concentrations in 2017 and 2018 remain unclear. Due to the fact that the most heavily contaminated sediments are still predominantly located in areas of the Middle Elbe, it can be assumed that the Hg concentrations in the harbors located there will only decrease very slowly. With regard to river maintenance and sediment management, it is likely that an improvement in sediment quality in the near future can only be achieved by remediating the harbor and groyne field areas and, in particular, by reducing the Hg loads of the Saale and Mulde rivers. Remediation of these two secondary sources of Hg, of which the Saale is to be regarded as more significant (Krüger et al., 2005; Baborowski & Einax, 2016), would also have a positive effect on the quality of SPM.

Spatial patterns of THg in sediments, FS, and SPM in the longitudinal course of the Elbe (2017–2019)

The samples from sediments, FS, and SPM differed in terms of grain size distribution, with sediments and SPM being dominated by silt and FS containing a higher average proportion of sand (Table 2). The percentages of TOC and N_t were on average lowest in FS and highest in SPM. Furthermore, the mean concentrations of S_t, Al, and Fe were highest in sediments and lowest in FS, while FS contained the highest percentage of Si, and SPM exhibited the highest concentrations of Mn. These differences must be considered when comparing the THg concentrations of the investigated sample groups. Especially, the differences

Table 2 Mean values, standard deviations (SD), minimal and maximal values of the grain size fractions sand (S), silt (U), and clay (C) (vol. %), the concentrations of total organic carbon (TOC), total nitrogen (N_t), C/N ratio, total sulfur (S_t), aluminum (Al), silicon (Si) (mass %), iron (Fe), manganese (Mn) (g kg^{-1}), total mercury (THg) (mg kg^{-1}), THg/TOC ratio (mg^{-1} kg TOC), and I_{geo} (Elbe background value 0.2 mg kg^{-1}) in sediments (SED) (2017–2019), floodplain soils (FS) (2020), and suspended particulate matter (SPM) (2019–2021)

Sample		Grain size (%)			TOC	N_t	C/N	S_t	Al	Si	Fe	Mn	THg	THg/TOC	I_{geo}
		S	U	C	(%)	(%)		(%)	(%)	(%)	(g kg^{-1})	(g kg^{-1})	(mg kg^{-1})		
SED (***n*** = 38)	Mean	34.8	61.8	3.4	5.7	0.5	10.5	0.35	4.48	24.6	34.1	1.34	2.06	40	2.2
	SD	18.8	17.8	3.0	1.7	0.2	1.6	0.17	1.19	3.98	8.49	0.63	2.05	32	1.4
	Min	10.4	16.1	0.1	1.1	0.1	7.8	0.06	1.89	16.1	9.87	0.31	0.11	2	−1.4
	Max	83.9	84.9	11.4	8.8	0.8	14.9	0.75	6.26	36.5	48.3	3.38	10.6	160	5.1
FS (***n*** = 25)	Mean	50.3	45.8	3.9	2.3	0.2	15.5	0.06	3.76	31.5	21.1	0.71	1.25	134	1.0
	SD	29.9	27.7	2.9	1.8	0.2	7.8	0.06	1.91	5.24	14.4	0.66	1.28	374	2.0
	Min	13.4	0.92	0.4	0.2	0.02	9.3	0.01	1.19	21.7	2.70	0.04	0.04	3	−2.8
	Max	98.5	78.3	10.8	5.7	0.61	35.0	0.31	6.33	38.2	49.0	2.54	4.20	1878	3.8
SPM (***n*** = 61)	Mean	37.6	57.0	5.5	5.9	0.7	9.1	0.23	4.02	21.5	34.0	3.75	0.88	13	1.4
	SD	12.2	11.9	1.9	1.2	0.2	1.4	0.09	0.98	4.37	7.73	1.44	0.43	6	0.7
	Min	21.8	28.7	3.4	2.4	0.2	6.2	0.09	1.67	12.5	15.6	0.81	0.37	7	0.3
	Max	66.3	72.8	9.1	7.8	1.1	11.6	0.44	5.29	31.5	57.0	8.50	1.84	26	2.6

in TOC concentrations explain why THg and I_{geo} reached the highest mean values in sediments, but the highest average THg/TOC ratio appeared in FS. We interpret this result to be due to the fact that half of the sampled FS comprise organic-poor subsoils that likely originate from older, comparatively more Hg-containing deposits. The comparison of THg concentrations using the Elbe-specific background value (0.2 mg kg^{-1}) leads to the classification of sediments to be "moderately to strongly contaminated" (I_{geo} 2.2), while FS and SPM are assessed to be "moderately contaminated." Regardless of this, the Elbe shows similar absolute concentration ranges as other highly Hg-contaminated rivers (Table S6). However, in comparison to global compilations of THg levels in rivers worldwide (e.g., Liu et al., 2021; Peng et al., 2025), in sediments of other European rivers (Salminen, 2005; Dendievel et al., 2022), or in European topsoils (Ballabio et al., 2021), the THg concentrations found in the Elbe river catchment are highly elevated and will likely remain so for a long time.

In accordance with the long-term monitoring sites, the concentration of THg in the longitudinal course of the Elbe significantly increases downriver of the confluences of the Mulde (river-km 260) and Saale (river-km 291) rivers within all sample groups (Fig. 3a). Furthermore, from river kilometer 260 to 585 the THg concentrations in SPM displayed lower variations than in sediments and FS, which indicates a smaller influence by Hg-enriched legacy deposits under average discharge conditions. Importantly, THg concentrations in sediments do not (yet) correspond to the lower THg concentrations of SPM, which is currently transported through the Elbe. Since the majority of the sediments and FS of the Middle Elbe are highly Hg-enriched, and Hg inputs via Saale and Mulde are still continuing, the deposition of Hg-enriched sediments will pose a persistent problem for the future maintenance of harbor basins and waterways in this region.

Although along the length of the Elbe the highest concentrations of THg were observed in sediments (Fig. 3b), the highest median value of THg/TOC ratio was found for FS (Fig. 3c), whose high variability can be explained by depth-specific differences of TOC in soil profiles. In particular, at a depth of 50–60 cm, a high variability of the Hg/TOC ratio was found, presumably due to the presence of older, Hg-enriched deposits and a comparatively advanced degradation of organic matter (Table S7). During the interpretation of both figures, it should be noted that the results exemplarily presented on FS originate from only 13 selected soil sampling sites situated in the immediate vicinity of the Elbe. Since significant changes of Hg concentrations can occur with changes in topography and floodplain morphology (Schulz-Zunkel &

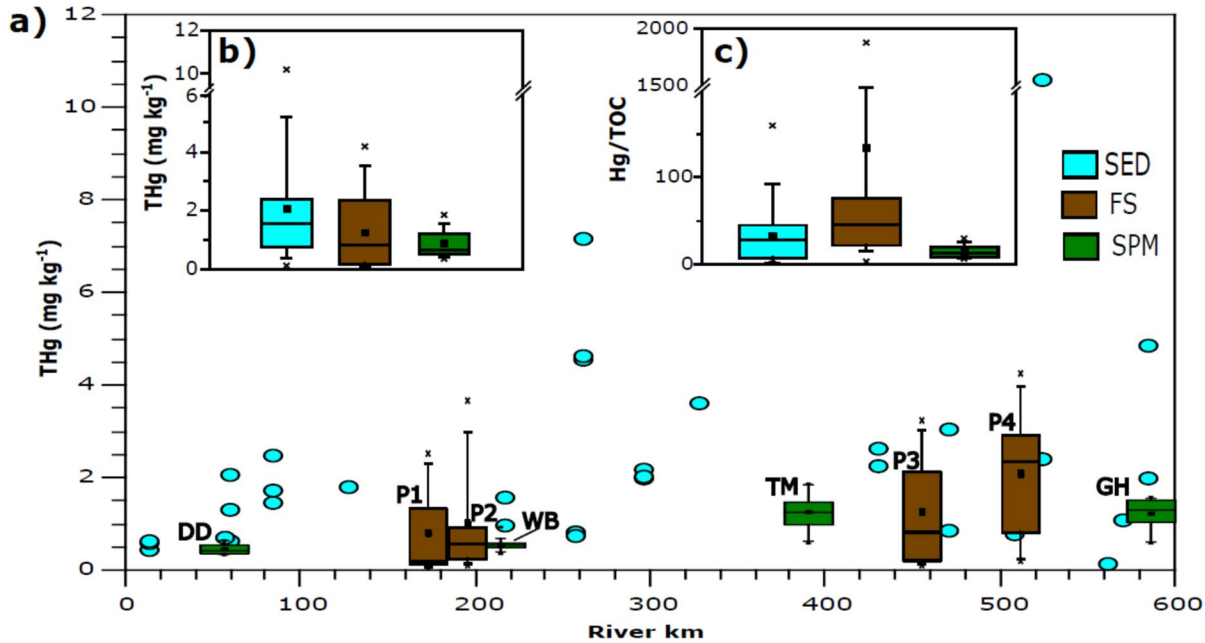

Fig. 3 (**a**) Spatial variations of THg concentrations in sediments (2017–2019, SED, sampling sites 1–18), FS plots P1–P4 (each boxplot comprises all samples of a soil plot), and SPM at the sampling stations Dresden (DD), Wittenberg (WB), Tangermünde (TM), and Geesthacht (GH) (each boxplot comprises all samples of a sampling station) in the longitudinal course of the Elbe. (**b**) THg concentrations and (**c**) THg/TOC (reported as mg Hg per kg TOC) ratio in all samples of sediments (2017–2019), FS, and SPM. Please note the axis breaks in panels (**b** and **c**)

Krüger, 2009), as well as between different soil types and soil horizons (Overesch et al., 2007; Du Laing et al., 2009), it is likely that the variability of THg in FS would be higher if a larger number of FS had been studied at different locations. For instance, a larger variability of THg concentrations in Elbe soils was published by Krüger and Gröngröft (2003).

Results of sequential extractions on Hg binding forms in different sample types

With respect to Hg binding forms studied by sequential extraction, F4 (HNO_3-extractable Hg) was the dominant binding fraction in sediments, followed by F3 (KOH-extractable Hg) and F5 (aqua-regia extractable Hg) (Fig. 4a, b). Clearly lower percentages occurred for F1 (water-extractable Hg) and F2 (extractable at pH 2), the fraction with the lowest proportions, which indicates a relatively strongly bound and little mobilizable state of Hg in sediments. These findings partially stand in contrast to the investigations by Wallschläger et al. (1998a), who found by means of a largely similar sequential extraction procedure a dominance of base-extractable Hg (corresponding to F3) in Elbe sediments. However, Wallschläger et al. (1998a) took their sediment samples directly from the stream bed and thus analyzed sediments with far lower concentrations of THg and mostly coarser grain sizes than investigated in this study, which highlights the importance of groyne fields and harbors as a sink of Hg within the Elbe region. In comparison, we assume that especially the occurrence of far higher THg in sediments explains the dominance of the F4 fraction in this study, which is in accordance with investigations by Schwab et al. (2022). Since for sediments a significant positive correlation was found between THg and F4 (r_s 0.63) and, in contrast, THg and F3 were negatively correlated (r_s −0.60) (Table 3), it appears that stronger THg enriched samples are especially dominated by Hg forms extracted in F4 in Elbe sediments. This effect is additionally visible when comparing the F4 percentage in samples between river km 0–260 and km 260–585 (i.e., up- and downriver of the Mulde River mouth), where in parallel to increasing concentrations of THg a significant increase in the percentage

Table 3 Spearman's rank correlation coefficients for THg and the concentrations of TOC, S_t, and the Hg binding fractions F1–F5, and the percentage of F1–F5 in sediments (SED) (2017–2019), FS (2020), and SPM (2019–2021) sampled along the longitudinal course of the Elbe (significant correlations are marked in bold, $^{**}p<0.01$; $^{*}p<0.05$)

	TOC	S_t	F1 (conc.)	F2 (conc.)	F3 (conc.)	F4 (conc.)	F5 (conc.)	F1 (%)	F2 (%)	F3 (%)	F4 (%)	F5 (%)
SED_THg	0.10	**0.76****	**0.66****	**0.62****	**0.57****	**0.97****	**0.78****	−0.01	0.02	**−0.60****	**0.63****	**−0.53****
FS_THg	**0.42****	**0.82****	**0.67****	**0.62****	0.14	**0.88****	**0.81****	−0.33	**−0.60***	−0.34	0.42	0.20
SPM_THg	**0.37****	**0.55****	**0.61****	**0.37***	0.31	**0.94****	**0.46***	0.20	0.08	**−0.83****	**0.81****	−0.23

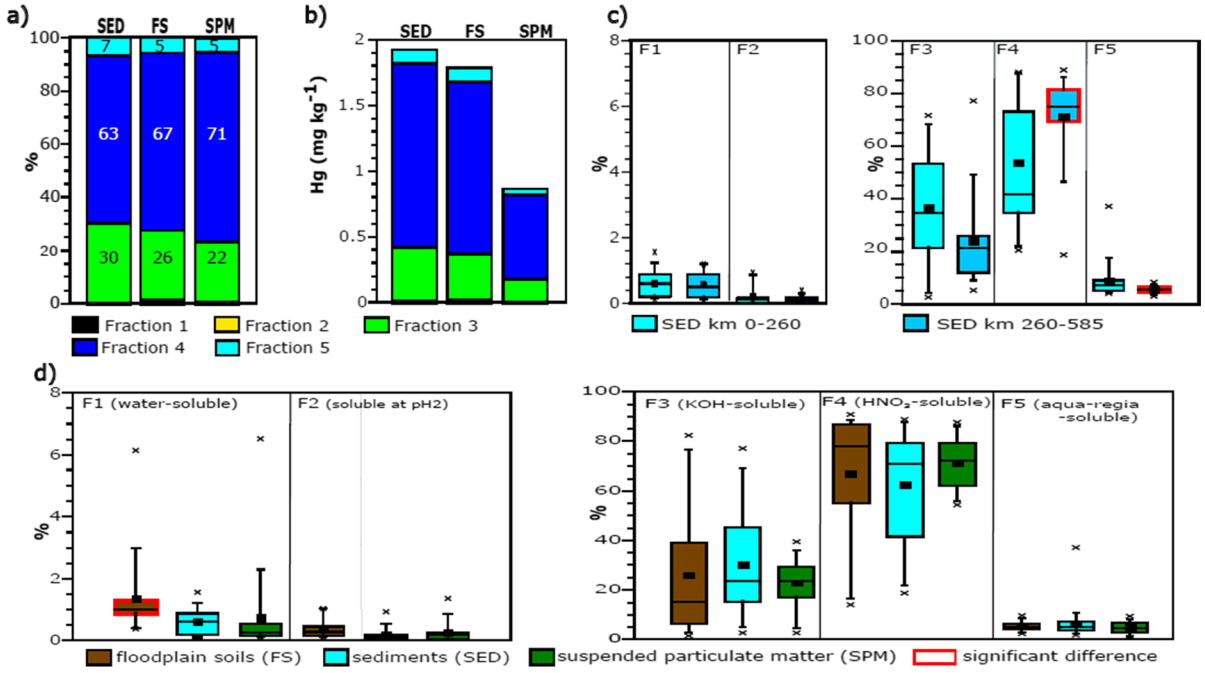

Fig. 4 (**a**) The mean percentages and (**b**) the mean concentrations of the Hg fractions in sediments (SED), floodplain soils (FS), and suspended particulate matter (SPM). (**c**) Boxplot diagrams of the binding fractions F1–F5 in sediments within the river transects 0–260 km and 260–585 km. (**d**) Comparison of the percentage of the Hg fractions F1–F5 in sediments (2017–2019), FS, and SPM. Boxes which are significantly different from the other boxes of a respective Hg fraction are marked red

of F4 was found (Fig. 4c, Table S8). In a comparison of Hg binding forms in sediments, FS, and SPM of the Elbe (Fig. 4d), it becomes evident that Hg in all investigated sample groups is dominated by F4. For FS, this finding agrees with the study by Frohne and Rinklebe (2013), who found a dominance of F4 in FS of the Elbe and Wupper Rivers. However, the relatively low mobility of Hg, as suggested by the dominance of F4, does not mean that Hg cannot potentially enter the food chain, as methylation plays a decisive role in this process and a reliable prediction of the Hg availability for methylation (e.g., Jonsson et al., 2012; Ndu et al., 2018) is not directly feasible based on sequential extraction data.

When interpreting data from sequential extractions, it should be noted that these are not able to describe the exact state of Hg binding in a sample (Shi et al., 2005). According to Bloom et al. (2003), F4 mainly comprises elemental Hg^0, which is the primary Hg-species when Hg is used as a catalyst in the chlor-alkali process (Wiederhold et al., 2015) and in other industrial applications (Brocza et al., 2019). Since the tributaries

Saale and Mulde were heavily impacted by historic chlor-alkali and acetaldehyde production (Hintelmann & Wilken, 1995), Hg could theoretically have entered the Elbe to high proportions as Hg^0, which could therefore in principle explain the dominance of F4 in sediments, FS, and SPM. However, considering the known stability of different Hg species under environmental conditions and the long time since most of the legacy inputs into the river system have occurred, it is rather unlikely that elemental Hg^0 still represents a relevant Hg form in the investigated samples. Thus, in contrast to the fraction assignment of Bloom et al. (2003), but in agreement with several other studies (e.g., Vasques et al., 2020; Covelli et al., 2021; Schwab et al., 2022), F4 may also be dominated by matrix-bound Hg^{2+} (i.e., Hg-complexes with organic matter, clay minerals, and metal(hydr-)oxides), as also demonstrated by Gilli et al. (2018). This is supported by the frequent occurrence of significant correlations between the concentrations of THg and TOC in sediments at the studied long-term sampling sites (Table S3). In contrast, samples from the longitudinal course of the Elbe were marked by a significant negative correlation between THg and F3 in sediments and SPM (Table 3). Therefore, it rather seems plausible that F4 contained significant proportions of organically bound Hg, which was either not completely extracted by F3 or which could be re-adsorbed on particle surfaces after it was released from organic substances by KOH. It is likely that re-adsorbed Hg of F3 would be subsequently extracted by F4. Furthermore, it cannot be excluded that fine-grained or poorly crystalline HgS compounds were already partly extracted in F4, as reported by Hall et al. (2005).

Significant differences in the percentages of binding fractions between sediments, FS, and SPM were only found for the fraction F1 (Fig. 4d, Table S9), which was significantly higher in FS than in other sample groups. This can be explained by the fact that water-soluble Hg in sediments and SPM is subject to continuous natural leaching by the river water, which reduces the levels of F1 (Kocman et al., 2004). Another remarkable feature, although not significant, is the occurrence of the comparatively greatest variability of fractions F3 and F4 in FS. This can be attributed to the high variability in grain sizes and strong differences of TOC and THg concentrations between the individual depth levels of the soils. F5 reached the largest range in sediments, which may be related to the fact that the formation of stable sulfidic compounds under anoxic conditions (Du Laing et al., 2009; O'Connor et al., 2019) may be favored in fine grained and organic rich sediments compared to other sample groups. In summary, significant differences in Hg binding, albeit to a minor extent, occurred both in sediments along the length of the Elbe as well as between the sample types sediments, FS, and SPM, so that hypothesis 2 could also be confirmed.

Results of correlation analyses

The results of correlation analyses of the samples taken from the longitudinal course of the Elbe (Table 3, Table S10) should be interpreted with caution. Since THg in sediments shows a gradient in its concentrations along the Elbe due to the inputs from the Saale and Mulde, random correlations with sediment characteristics that are also subject to a gradient in the longitudinal course are favored, as mentioned for S_t above. Conversely, correlations with TOC, for which a relationship to the concentration of THg is typical (Heise et al., 2008), are likely suppressed within the evaluated data set since it is not subject to a gradient along the Elbe.

However, significant correlation coefficients show that the absolute concentrations of the fractions F1, F2, F4, and F5 (in sediments also F3) increase with increasing THg, independent of a respective sample group. Since THg and the absolute Hg concentrations in extracted fractions are not independent parameters, such correlations seem rather obvious. However, with regard to the fractions F1 and F2, this means that Hg is increasingly mobilized under increasing THg and thus the potential of mobile Hg entering organisms and the food chain is significantly enhanced at elevated THg levels. Since fine sediments also accumulate nutrients and often exhibit strongly anaerobic conditions, THg enrichment could additionally favor the microbial production of highly toxic methylmercury, as Hintelmann and Wilken (1995) already demonstrated for Elbe sediments and which is also a subject of an ongoing research project (Wiederhold et al., 2024).

The percentages of the binding fractions in sediments and SPM are positively correlated with THg in the case of F4 and negatively correlated in the case of F3. This is due to the fact that with increasing THg concentration, increasing amounts of HNO_3-soluble Hg were extracted, whereby the percentages of other

Hg fractions are partially reduced (see above for a discussion of the causes). Surprisingly, this phenomenon appears to be less pronounced in FS, which could be due to differences in the soil types of the investigated floodplain sites.

Conclusions

The investigations show that the temporal development of THg concentrations between 1996 and 2019 in sediments of the Elbe is highly dependent on the location of the five studied long-term monitoring sites. Sediment samples taken from the longitudinal course of the Elbe (2017–2019) showed the highest THg concentrations in areas of the Middle Elbe, with sediments and FS exhibiting significantly higher concentrations of THg than SPM. Our major findings are as follows:

- Time series of Hg concentrations in fine sediments of harbors and groyne fields only partially decreased during the recent past. Especially sediments collected at sites of the middle Elbe did not show declines in THg and did not reflect the declining concentrations of THg in SPM.
- A time lag still exists between the decline in Hg releases into the Elbe and the reduction in Hg concentrations in the fine sediments of the river course.
- Due to the very slow change in the Hg contamination status, Hg-enriched deposits will continue to pose a future problem for the management of harbor basins and waterways (e.g., Hsu-Kim et al., 2018; Eckley et al., 2020; Ma et al., 2023; Amyot et al., 2025).
- Sequential extractions revealed that Hg in sediments, FS, and SPM is dominated by strongly bound Hg forms, suggesting a relatively low THg mobility.

However, even if the relative proportion of easily mobilizable Hg compounds is low, there is a risk of microbially catalyzed conversion of inorganic Hg into highly toxic organic methyl-Hg compounds under suitable biogeochemical conditions in localized riverine environments (Bigham et al., 2016; Singer et al., 2016; Krause et al., 2024). Since these accumulate in organisms along the aquatic food chain, the environmental quality standard of the EU Water Framework Directive for Hg in biota is likely to continue being exceeded even in the future decades.

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Author contribution JH: Conceptualization, Methodology, Investigation, Writing-Original Draft LB: Methodology JW: Writing-Original Draft, Writing-Review and Editing.

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Data availability All data are found in tables or figures within the manuscript, the supplementary information (SI) and at https://zenodo.org/records/17897002. If further information is required, it can be requested from the corresponding author.

Declarations

Ethics approval All authors have read, understood, and have complied as applicable with the statement on "Ethical responsibilities of Authors" as found in the Instructions for Authors.

Competing interests The authors declare no competing interests.

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